

■ Improving Urban Air Quality with AI (SDG 11: Sustainable Cities and Communities)

1. The Problem

Air pollution is one of the most pressing challenges faced by growing cities today. Rapid urbanization, industrial activities, and traffic congestion have made it difficult for city authorities to track and manage air quality effectively. According to the UN, nearly 99% of the global population breathes air that exceeds WHO pollution limits, contributing to millions of premature deaths each year. In line with Sustainable Development Goal 11 (SDG 11) — “Make cities inclusive, safe, resilient, and sustainable” — this project explores how artificial intelligence can help identify pollution patterns and support data-driven environmental planning.

2. The Approach

To understand air quality across different U.S. cities, this project uses a machine learning clustering approach. Specifically, it applies the K-Means algorithm, an unsupervised learning method that groups cities based on similarities in key air pollutants: - Nitrogen Dioxide (NO₂) - Ozone (O₃) - Sulfur Dioxide (SO₂) - Carbon Monoxide (CO) By standardizing these pollutant levels and clustering the data, the model reveals groups of cities that share similar pollution profiles — from relatively clean air to highly polluted urban zones.

3. The Data and Tools

The dataset comes from Kaggle's U.S. Pollution Data, which provides pollutant averages across major cities. Tools used: Python, Pandas, Scikit-learn, Matplotlib, and NumPy Workflow steps: data cleaning, normalization, clustering, and visualization For GitHub sharing, a smaller dataset (pollution_sample.csv) with 500 rows is included to ensure reproducibility and fast runtime.

4. The Findings

The K-Means model grouped cities into four main clusters: - Cluster 0: Clean-air cities with low levels of all pollutants - Cluster 1: Cities with moderate ozone and nitrogen dioxide - Cluster 2: Industrial regions with high sulfur and carbon monoxide - Cluster 3: High-pollution clusters that require urgent air quality management A 2D scatter plot of NO₂ vs. O₃ clearly illustrates how urban centers differ in their air composition — offering a visual tool for environmental policy analysis.

5. Ethical and Social Reflection

While clustering helps uncover pollution patterns, it's crucial to remember that datasets may not represent all communities equally. Some regions might lack complete air quality data, which could bias results. Ensuring fair data coverage across all cities helps avoid underrepresenting vulnerable populations. This project promotes fairness and sustainability by making air quality information

more transparent and actionable, empowering city planners and policymakers to focus interventions where they are most needed.

6. Conclusion

By combining open environmental data and AI, this project demonstrates how technology can support SDG 11 goals. Machine learning enables us to see patterns that might otherwise be hidden — guiding cleaner, healthier, and more resilient urban environments for everyone.